



Peripheral Intravenous Cannula (PIVC) Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Health Service	All areas	Medical; Nursing

2. Purpose

This document provides guidance for clinicians to provide optimal care for patients having a peripherally intravenous cannula (PIVC) inserted. It describes the requirements for the insertion and management of PIVC to minimise the risk of infection to patients associated with the insertion of a PIVC. This document is based on the Communicable Disease Control Directorate Guideline for Insertion and Management of Peripheral Intravenous Cannulae in Healthcare Facilities.

3. Insertion and management of PIVC

Only Healthcare Workers (HCW) who are appropriately trained and deemed competent in this procedure should insert peripheral venous cannulae.

At AKG, Nursing & Allied Health staff are deemed competent when they have completed the Learning Management System (LMS) PIVC package. This includes e-learning, attending a face to face PIVC insertion workshop and completing two successful PIVC insertion competency assessments which need to be uploaded to the LMS.

A register of competent Nursing and Allied Health staff is kept in the LMS.

Medical staff are deemed competent when they have completed an EMHS/RPH PIVC competency workshop which should be recorded in the LMS.

3.1 Pre-insertion considerations

- A PIVC is only to be inserted if deemed clinically necessary and other alternatives are not an option, e.g. oral medication
- The DRIP criteria should be used to identify patients for whom cannulation could be eliminated or delayed until a definitive indication for a PIVC is established. PIVCs should also be removed when no longer indicated by the DRIP criteria. The DRIP criteria indicating PIVC insertion are as follows:
 - D** – Deteriorating / risk of sudden deterioration
 - R** – Rehydration: IV fluids prescribed
 - I** – IV Medications prescribed Site Operational Area Applicable to
 - P** – Procedures requiring cannulation indicated

PIVC should not be inserted into the cubital fossa except in an emergency situation or where large bore access is required.

- If venous access is required, ensure the most appropriate venous access device is chosen (See appendix 1) e.g. when repeated or prolonged administration of vesicant or irritant solutions, such as potassium chloride, flucloxacillin or vancomycin, is required, central venous access should be considered, to avoid peripheral vein damage.⁹
- See appendix 2 for PIVC site and size selection.
- The use of local anaesthetic, such as subcutaneous lignocaine or EMLA cream to reduce the pain of insertion, should be considered before the insertion of any PIVC, regardless of PIVC size and age of patient unless contraindicated e.g. use of the PIVC for vesicants or irritant solutions.¹⁰⁻¹²
- EMLA cream can leave a lipid residue that may create a focus for microbial growth therefore prior to skin decontamination residual topical anaesthesia should be removed with soap and water.¹³
- Prior to insertion, HCWs are to confirm patient identity and obtain verbal consent and advise patients their role and involve them in their care to prevent PIVC related complications.

3.2 Difficult IV access

- The HCW inserting the PIVC is to seek assistance from a more experienced HCW after two unsuccessful attempts (see Appendix 1, Difficult Intravenous Access Pathway for escalation protocol).
- If three or more attempts within 2 days, consider insertion of a midline or PICC (Appendix 1).
- Specific considerations for neonates and paediatric patients are outlined in section 12.

3.3 PIVC device selection

- The use of PIVC that are classed as safety-engineered medical devices (SEMDs) is preferred to reduce the risk of injury involving a sharp.⁷
- The size of the PIVC is to be determined by the intended use, e.g. hydration, blood products, the condition of the patient's veins and the insertion site. The PIVC is to be the shortest and smallest gauge that is suitable for the anticipated clinical need (refer to Appendix 2).¹⁴
- All PIVC are to have an extension set attached, e.g. j-loop. Extension sets help maintain stability and reduce trauma to the vein.⁸

3.4 PIVC site selection

- PIVC are to be routinely sited in the distal areas of the upper limbs. Subsequent
- PIVC are to be inserted, where possible, proximal to the previous site. Select the
- most appropriate vein for insertion of the PIVC with consideration of:
 - Indication and expected duration of PIVC.
 - Size and condition of patient's veins.
 - Position of patient during any planned procedure(s).
 - Utilising the patient's non-dominant forearm if practical.
 - Utilising basilic or cephalic veins on the posterior (dorsal) forearm if possible.

If possible, avoid the use of veins in the following sites:

- The anterior (ventral) forearm veins, especially the cephalic vein, in patients with chronic renal failure, as these may be required for fistula formation for dialysis.

- Areas of flexion, e.g. antecubital fossa, or bony prominences due to increased risk of BSI and discomfort for the patient.
- Areas below previous cannulation sites bruised or phlebitic areas due to poor venous return and possibility of clots being dislodged.
- A limb with an arteriovenous fistulae or shunt, as this may compromise access for haemodialysis.
- An arm on the same side as a previous axillary clearance, mastectomy or affected by a cerebrovascular accident.
- An infected limb, e.g. with cellulitis, due to increased risk of infection.
- A limb with a PICC or implanted venous access device.
- Lower limbs due to risk of deep vein thrombosis, reduces access, patient comfort and mobility.¹²

3.5 Standard precautions and aseptic technique

- A risk assessment is to be carried out to identify the standard precautions required to safely perform the procedure.⁷
- Hand hygiene consistent with the '5 Moments for Hand Hygiene' with an antiseptic solution is to be performed immediately prior to the insertion of PIVC.⁷
- Aseptic technique is always to be utilised during the insertion and ongoing management of the PIVC.^{7, 16}
- A sterile PIVC starter pack or sterile dressing pack and associated sterile equipment are to be used to insert all PIVC.
- Sterile gloves are to be utilised for insertion, to assist in maintaining aseptic technique in case re-palpation of the insertion site following skin disinfection is required and while connecting the hub of the cannula to the extension set.

3.6 Skin preparation

- Use clippers to remove hair at the insertion site if necessary.
- Clean the skin with neutral soap and water if the insertion site is visibly soiled.
- Perform skin disinfection using 2% chlorhexidine gluconate in 70% isopropyl alcohol solution, except in the case of documented allergy or in neonates.⁷
- Liberally swab a large area of skin around the chosen insertion site to ensure the site for the dressing is also disinfected.
- Allow skin antiseptic to air dry to ensure adequate contact time. Do not wipe or blot skin dry.
- For patients with a history of chlorhexidine sensitivity / allergy, use povidone iodine 10% in 70% ethyl alcohol, and allow to air dry. If alcohol is contraindicated, use an aqueous 10% povidone-iodine solution.

3.7 Securement and dressing management

- Use a sterile, transparent semi-permeable dressing to secure the PIVC, extension set, or needleless valve if short stay device, to stabilise and secure the PIVC, allow continuous observation of the site and protect the insertion site from contamination.
- Secure the dressing, taking care not to contaminate the adhesive part of the dressing, where the cannula hub and the extension set connect and ensuring the dressing is firmly adhered to the skin.

- The insertion site should remain visible for inspection, therefore, do not place opaque tape directly over the insertion site.
- Record the date and time of insertion on the adhesive strip of the IV dressing and on the Peripheral Intravenous Cannula Form (AKMR121.6).
- The dressing is to be replaced using aseptic technique if it becomes wet, soiled or loose.
- If a PIVC becomes partially withdrawn or dislodged, the PIVC is to be removed and a new PIVC inserted as soon as practical 13.

4. PIVC assessment

PIVC sites and devices must be assessed at least once per shift or at eight hourly intervals or if clinically indicated.

- The PIVC and insertion site are assessed for signs of complications that can lead to device failure (refer to appendix 3)
- The PIVC should be removed as soon as it is no longer required.⁷
- All PIVCs are to be assessed for patency and for any signs of complication each time the device is accessed and determine if the patient is tolerating their PIVC and their understanding of the need for the device. 7
- More frequent assessment is indicated if there are:
 - signs of pain, swelling or redness at the insertion site, by visual inspection through the transparent dressing and gentle palpation through the dressing
 - the condition of the patient's veins, and whether they have become hardened or thrombosed
 - signs of localised or systemic infection (if either are confirmed, report in Datix CIMS)
 - leakage of fluid from the insertion site, signs of occlusion, infiltration or extravasation
 - concerns as to whether the PIVC remains appropriately dressed and secured.⁷
- All PIVC are to have a peripheral intravenous assessment score (PIVAS) performed each shift or at least every eight hours while the PIVC is insitu and continued for 48 hours post removal.
- Any PIVC site issues are to be documented in the patient's medical record (refer to Appendix 3).

5. PIVC blood collection

- Blood samples may only be drawn from a PIVC directly after insertion. Do not routinely aspirate blood samples directly from PIVC due to the potential risk of haemolysis and thrombus formation in the catheter tip. Emergency situations are the exception when the patient has limited vascular access or is at increased risk of bleeding or receiving thrombolytic therapy.^{13, 14, 17} The collection of blood cultures at time of insertion of a PIVC is not encouraged due to the increased risk of contamination at the time of collection. The exception is in neonates, infants and children (refer to section 12). If the procedure is performed, a second set of blood cultures collected by venepuncture are to be collected.^{17,}

6. Needleless access ports

As closed intravenous access systems are associated with fewer BSIs than open systems, needleless access ports are to be used on all lumens.⁹ Stopcocks are to be end-capped with a needleless access port when not in use.

- All PIVC access ports are to be disinfected by rubbing with a single-use 70% alcohol impregnated swab for 20-30 seconds and allowed to dry for at least 30 seconds prior to accessing the system. 70% alcohol has significant and immediate antimicrobial activity and reduces unnecessary exposure to chlorhexidine as the residual activity of chlorhexidine is not required on inanimate surfaces. 9, 15, 19, 20
- All access ports are only to be accessed with a sterile single-use device.
- When an access port is removed from a PIVC or extension set, it is to be discarded and a new sterile access port attached.

7. Management of administration sets

- Administration sets, including all tubing, connections, extension sets, and needleless valves are to be changed when the PIVC is re-sited at 72 hours.
- Administration sets are to be changed more frequently if contamination or accidental disconnection occurs or a blood reaction is suspected.
- When blood or blood products have been infused, change the administration set, including all IV tubing and connections immediately after completion of the infusion or every 12 hours, whichever comes first. 21
- Administration sets are single use devices and if they are disconnected from the intravenous cannula for any reason, e.g. intermittent medication dosing, the set is to be discarded and a new administration set connected using aseptic technique. 22
- Administration sets are not to be disconnected for routine care, e.g. showering, but may be disconnected for transient, controlled disconnections, e.g. changing IV access or infusions in operating theatres or medical imaging departments.
- Label all administration sets attached to the PIVC with an intravenous line label in accordance with the National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines. 23

8. PIVC flushing

- If the patient is receiving intermittent injections or infusions, flushing under positive pressure is recommended to promote and maintain patency and prevent the mixing of incompatible medications and solutions.
- PIVC are to be flushed with 5-9mls of sterile 0.9% sodium chloride for injection using a 10ml luer-lock syringe or commercially available pre-filled syringe to help avoid excessive pressure.
- HCWs are to flush PIVC, using a pulsatile motion (push-pause):
 - after the PIVC is inserted and prior to use to confirm placement
 - before each medication or infusion is given to ensure the PIVC is still patent
 - after each injection / infusion to remove irritant material from the vein
 - between multiple infusions or medications to prevent interactions and incompatibilities

- before and after blood drawing (refer to section 5) PIVC blood collection
- at least every 12 hours if the PIVC is not in use (strong consideration should be given to removing the PIVC if it has not been accessed for 12 hours).
- Disconnecting the flush syringe can allow reflux of blood into the tip of the catheter to displace the space occupied by the syringe. To prevent this source of occlusion, HCWs must clamp the extension set before disconnection or withdraw the syringe while administering the last 0.5 ml of flush (positive pressure technique).

9. Duration and re-siting of PIVCs

- All PIVC inserted into the cubital fossa are to be reviewed within 24 hours and resited if clinically indicated.
- If the reviewing doctor decides that the PIVC **MUST** stay in the cubital fossa, this must be documented in the DMR and noted as discussed with a registrar as a minimum regarding infection risk associated with leaving it in place
- All PIVC inserted by ambulance or other emergency services are to be reviewed as soon as the patient's condition has stabilized or within 24 hours of insertion and replaced if clinically indicated. 7
- Remove PIVCs that may have been inserted without adherence to aseptic technique e.g. resuscitation as soon as practical and within 24 hours of insertion.
- Any patient with a PIVC following inter-hospital transfer, is to have their PIVC assessed for clinical need, the time in situ and a PIVAS assessment undertaken. Actions should be directed based on findings. If unable to determine adequate information the PIVC should be removed and re-sited if ongoing need.
- All PIVC are to be reviewed daily or when clinically indicated, for ongoing need and removed as soon as no longer required (see DRIP criteria) and are not to remain in situ longer than 72 hours. 9, 14, 16, 20

Patients with known or anticipated difficult intravenous access (DIVA) should be referred to the Vascular Access Clinical Nurse Specialist as soon as possible before the 72 hour mark.

- The responsible medical officer is to review the need for PIVC access daily, and if ongoing access is required past 72 hours, planned re-siting of the PIVC is to occur.
- Remove the PIVC if PIVAS is 2 or greater, or fever $>38^{\circ}\text{C}$, or signs of sepsis are evident.
- If prolonged IV therapy is likely to be required, consideration for a central catheter, either peripherally or centrally inserted, or a long peripheral venous catheter (also referred to as a 'midline' catheters) should be utilised rather than multiple replacement of PIVCs (see appendix 2 for choosing appropriate venous access on the ward).

9.1 Removing the PIVC if infection suspected

- If infection is suspected, inform the treating medical officer. Two sets of blood cultures are to be collected. Blood culture samples are to be drawn from another peripheral vein. Blood must not be drawn from the existing PIVC. Ensure aseptic technique occurs during sampling.
 - Any PIVC site discharge should be swabbed and sent for culture.
 - On removal of the PIVC send catheter tip for culture in a sterile screw top container. NB: blood cultures must accompany tip.

- All actions are to be documented in the patient's medical record. Report significant local and PIVC related site infection on Datix CIMS, in accordance with MP 0122/19 Clinical Incident Management Policy.²⁴

10. Documentation requirements for PIVC

- All documentation in relation to a PIVC is to be recorded as part of the patient's medical record and maintained as a permanent record using the Peripheral Intravenous Cannula Form (AKMR121.6).
- For each PIVC inserted, the documentation is to include:
 - date and time of insertion
 - anatomical site of insertion
 - name of the HCW inserting the PIVC
 - removal date and time
 - reason for removal e.g. treatment complete, pain, dislodgement, PIVAS 2 or greater, extravasation, vessel hardness or emergency insertion
- Documentation is to include details of any PIVC that has been inserted in emergency situations or when there have been failed insertion attempts. This can be documented in dedicated emergency PIVC section in AKMR 121.6.
- The use of an IV insertion label, noting date of insertion is to be used as a visual prompt on the PIVC dressing. They are to be placed on the external transparent dressing, so that they are visible but will not interfere with viewing the PIVC site.
- A PIVAS is to be recorded for each PIVC site, each shift for the duration the PIVC is in situ and for 48 hours following removal to detect post removal complications (Refer Appendix 3). Ongoing PIVC site issues beyond 48 hours are to be documented in the patients' medical record.
- All clinical interventions for each PIVC site are to be recorded in the patients' medical record.

11. Patient education

- Ensure patients or carers are provided with information in relation to their PIVC and possible complications (refer to IPPSU tools and resources webpage). HCWs are to have a discussion with the patient, when mentally competent, to ensure they understand the information provided to them.
- Document on AKMR 121.6 tick-box that information has been provided.

12. Neonate and Paediatric Consideration

12.1 Definitions

• Neonate: less than 28 days old	• Infant: 1 month to 12 months old
• Preterm neonate < 37 weeks gestational age	• Child: for the purpose of this document – • refers to children aged 12 months to 16 years.

12.2 PIVC site preferences

It is preferable to utilise veins in the hands and feet and to choose veins that run straight, fill and empty and are easy to splint, although sites such as the scalp in neonates and infants can also be used.

12.3 Insertion

Neonates	Infants, children and adolescents
- Standard tourniquets are not to be used. - Occlusion of the vein can be achieved with gentle pressure applied to the vein proximal to the insertion site. - Safety-engineered medical devices are generally not used due to the complexity and difficulty of the procedure. - A 24g cannula is to be utilised.	- A single-patient use tourniquet of the appropriate size is recommended. - Safety-engineered medical devices, where available, are recommended to reduce the risk of sharps injury.

12.4 Skin disinfection

Neonates	Infants, children, and adolescents
- If ≤ 28 weeks gestation utilise povidone-iodine 10% solution and allow to air dry. - Remove antiseptic solution with sterile saline or sterile water before proceeding with the procedure. - It is recommended to avoid the use of chlorhexidine in extremely low birth weight babies due to the risk of chemical burns.^{25, 26} - If > 28 weeks gestation utilise 1% chlorhexidine solution and allow to air dry before proceeding with the procedure. - Wash off excess solution after the procedure with sterile water or saline to prevent chemical burns.^{25, 26}	Perform skin disinfection using 2% chlorhexidine gluconate in 70% isopropyl alcohol solution, except in the case of documented allergy.⁷

12.5 Pain Management

Neonates and infants	Children and adolescents
Non-pharmacological interventions have been shown to be effective for pain management in Neonates and Infants	Utilise distraction, play therapy, topical anaesthetic, 'Buzzy' device, relaxation, breathing and imagery techniques in accordance with the child's developmental

- Recommended practices may include breastfeeding, skin to skin contact, oral sucrose, breast milk, kangaroo care, holding, swaddling, oral sucrose and non-nutritive sucking.^{27, 28} - Repeated use of sucrose should be used with caution in pre-term infants less than 31 weeks gestation.²⁹	stage and considering previous experiences and anxieties.
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12.6 Securement and dressing and management

- The type of securement for the PIVC depends upon several factors, including the condition of the skin, site of the PIVC, mobility of the neonate / child and risk of dermal stripping.
- When dressing the PIVC ensure the dressing is secure, the site is visible and that the taping is not occlusive or restrictive.
- IV boards or splints are recommended to secure PIVC placed in or adjacent to areas of flexion, to immobilise the joint and minimise the risk of venous damage.
- When using a splint, ensure it is positioned and strapped with the limb and digits in a neutral position and the taping is not occlusive or restricting circulation.
- If securing the splint with tape, consider lightly backing any tape with cotton wool or gauze that has contact with skin.
- Consider placing a small piece of cotton wool ball or gauze underneath the hub of the cannula to reduce risk of pressure injury.
- Inspect the splint each shift and change if wet or soiled.

12.7 Flushing

- Use pre-prepared 0.9% sodium chloride flush syringes e.g. Posiflush in preference to drawing up sodium chloride with a syringe and needle.
- For neonates, infants and children, use the minimum volume of flush to clear a line and any add-on devices, of fluid, medication or blood (between and after each medication or fluid administration), as follows: 30
 - Neonates: 0.5mL
 - Infant: 2ml
 - Child / Adolescent: 5 - 10mL.
- The IV device should be flushed using pulsatile positive pressure technique.
- PIVCs without a continuous infusion are to be flushed 6 to 8 hourly to ensure patency.

12.8 PIVC assessment

- Inspect the PIVC insertion site at least hourly when a continuous infusion is in progress, and with each intermittent medication and flush administration, ensuring any covering is removed completely to perform an assessment of the insertion site and to observe the limb above and below the site.
- Any adverse findings are to be documented in the patient's medical record.
- PIVAS documentation is to be applied in the neonatal and paediatric settings.
- Increased supervision is required for active infants/young children on continuous infusions due to the risk of entanglement with administration lines.

12.9 PIVC blood collection

- Blood samples, including blood cultures, may be drawn from a PIVC directly after insertion, but not at other times.
- For neonates, regular routine blood sampling following PIVC insertion depends on sample volume required and should be via capillary heel prick (for volumes <1ml) or venepuncture (for volumes >1ml). An arterial line should be used for critically ill neonates requiring frequent blood sampling. 31
- In infants and children, if multiple blood samples are required for short term investigative procedures or emergency management a peripheral blood sampling line can be inserted. 32

12.10 Duration of PIVC

- PIVC related infections are less prevalent in children than in adults, and due to the difficulty in establishing intravenous access in this population PIVC's are not routinely replaced.
- The PIVC can stay in situ if:
 - a PIVC is clinically indicated
 - there is no evidence of local (redness, pain, tracking) or systemic (fever and rigors) signs of infection is still flushing well without resistance or leakage from the insertion site

12.11 Management of administration sets

- Administration sets, including all tubing, connections, extensions sets and needleless valves are to be changed when the PIVC is re-sited, if contamination or accidental disconnection occurs or a blood reaction is suspected. 9, 32

13. Compliance monitoring

Compliance with the requirements of this procedure will be monitored by the Infection Prevention and Management Committee.

Evaluation of this procedure will be carried out by the following means:

- Infection Prevention and Management (IPM) team will review and report all identified healthcare associated *Staphylococcus aureus* blood stream infections (HA-SABSI) related to PIVC in as per [Reporting of healthcare-associated Staphylococcus aureus bloodstream infections as a severity assessment code 1\(SAC1\)](#) policy.
- IPM committee will review reported clinical incident data related to PIVC associated issues and any corrective actions that have been undertaken, as reported to them.
- Scheduled retrospective clinical audits to monitor documentation in relation to PIVC selection, insertion, monitoring and removal.

14. Policy context

The following documents are required to give effect to this procedure:

- [Communicable Disease Control Directorate Guideline Insertion and Management of Peripheral Intravenous Cannulae in Healthcare Facilities](#)
- [Insertion and Management of Peripheral Intravenous Cannulae in Healthcare Facilities Policy MP 0038/16](#)

15. Supporting information

The following documents support the implementation of this procedure:

- [Management of Peripheral Intravenous Catheters Clinical Care Standard National Safety and Quality Health Service Standards. Australian Commission on Safety and Quality in Healthcare \(ACSQHC\) \(2021\)](#)

16. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.3
- Standard 3 – Action 3.9
- Standard 3 – Action 3.10

17. References

1. Chopra V, Saint S. Vascular catheter infections: time to get technical. *The Lancet*. 2015;386(10008):2034-6.
2. Rhodes D, Cheng AC, McLellan S, Guerra P, Karanfilovska D, Aitchison S, et al. Reducing *Staphylococcus aureus* bloodstream infections associated with peripheral intravenous cannulae: successful implementation of a care bundle at a large Australian health service. *J Hosp Infect*. 2016;94(1):86-91.
3. Collignon PJ, Dreimanis DE, Beckingham WD, Roberts JL, Gardner A. Intravascular catheter bloodstream infections: an effective and sustained hospital-wide prevention program over 8 years. *Med J Aust*. 2007;187(10):551-4.
4. Turnidge JD, Kotsanas D, Munckhof W, Roberts S, Bennett CM, Nimmo GR, et al. *Staphylococcus aureus* bacteraemia: a major cause of mortality in Australia and New Zealand. *Med J Aust*. 2009;191(7):368-73.
5. Healthcare Associated Infection Unit. Healthcare Infection Surveillance Western Australia (HISWA) Annual Report 2017-18. Western Australia: Department of Health; 2019.
6. Patient Safety and Clinical Quality Division. Western Australian Safety and Quality Point Prevalence Survey 2014. Western Australia: Department of Health 2014.
7. Australian Commission on Safety and Quality on Health Care. Management of Peripheral Intravenous Catheters Clinical Care Standard NSW: ACSQHC; 2021 [1 August 2022]. Available from: https://www.safetyandquality.gov.au/sites/default/files/2021-05/management_of_peripheral_intravenous_catheters_clinical_care_standard_-_accessible_pdf.pdf.
8. Australian Commission on Safety and Quality on Health Care. 5 Moments For Handy Hygiene NSW: ACSQHC; 2022 [1 August 2022]. Available from: <https://www.safetyandquality.gov.au/our-work/infection-prevention-and-control/nationalhand-hygiene-initiative-nhhi/what-hand-hygiene/5-moments-hand-hygiene>.
9. O'Grady NP, Alexander M, Burns LA, Dellinger EP, Garland J, Heard SO, et al. Guidelines for the prevention of intravascular catheter-related infections. *Am J Infect Control*. 2011;39(4 Suppl 1): S1-34.

10. Tee FY, Low CSL, Matizha P. Patient Perceptions and Experience of Pain, Anxiety and Comfort during Peripheral Intravenous Cannulation in Medical Wards: Topical Anaesthesia, Effective Communication, and Empowerment. *International Journal of Nursing*. 2015;5:41-6.
11. Koh JL, Harrison D, Myers R, Dembinski R, Turner H, McGraw T. A randomized, double-blind comparison study of EMLA and ELA-Max for topical anesthesia in children undergoing intravenous insertion. *Paediatr Anaesth*. 2004;14(12):977-82.
12. Harris T, Cameron PA, Ugoni A. The use of pre-cannulation local anaesthetic and factors affecting pain perception in the emergency department setting. *Emerg Med J*. 2001;18(3):175-7.
13. Queensland Health. Intra-vascular device management: Queensland Government; 2019 [15 September 2022]. Available from: <https://www.health.qld.gov.au/clinical-21> Guideline: Insertion and Management of Peripheral Intravenous Cannulae in Healthcare Facilities practice/guidelines-procedures/diseases-infection/infection-prevention/intravasculardevice-management.
14. New South Wales Health. Peripheral Intravenous Cannulation (PIVC) Insertion, Care and Removal (Adults) NSW Government: South Eastern Sydney Local Health District; 2022 [15 September 2022]. Available from: <https://www.seslhd.health.nsw.gov.au/node/9255>.
15. Loveday HP, Wilson JA, Pratt RJ, Golsorkhi M, Tingle A, Bak A, et al. epic3: national evidence-based guidelines for preventing healthcare-associated infections in NHS hospitals in England. *J Hosp Infect*. 2014;86 Suppl 1:S1-70.
16. National Health and Medical Research Council. Australian Guidelines for the Prevention and Control of Infection in Healthcare: Australian Commission on Safety and Quality in Health Care; 2019 [15 September 2022]. Available from: <https://www.nhmrc.gov.au/about-us/publications/australian-guidelines-prevention-and-control-infection-healthcare-2019#block-views-block-file-attachments-content-block-1>.
17. Self WH, Speroff T, McNaughton CD, Wright PW, Miller G, Johnson JG, et al. Blood culture collection through peripheral intravenous catheters increases the risk of specimen contamination among adult emergency department patients. *Infect Control Hosp Epidemiol*. 2012;33(5):524-6.
18. Bentley J, Thakore S, Muir L, Baird A, Lee J. A change of culture: reducing blood culture contamination rates in an Emergency Department. *BMJ Qual Improv Rep*. 2016;5(1).
19. Maiwald M, Chan ES. The forgotten role of alcohol: a systematic review and metaanalysis of the clinical efficacy and perceived role of chlorhexidine in skin antisepsis. *PLoS One*. 2012;7(9):e44277.
20. Stuart RL, Cameron DRM, Scott C, Kotsanas D, Grayson ML, Korman TM, et al. Peripheral intravenous catheter-associated *Staphylococcus aureus* bacteraemia: more than 5 years of prospective data from two tertiary health services. *Medical Journal of Australia*. 2013;198(10):551-3.
21. Australian & New Zealand Society of Blood Transfusion Ltd. Guidelines for the Administration of Blood Products: National Blood Authority Australia; 2018 [4 May 2022]. Available from: <https://www.blood.gov.au/guidelines-administration-blood-products>.
22. Therapeutic Goods Administration (TGA). Australian regulatory guidelines for medical devices (ARGMD): Commonwealth of Australia; 2022 [23 August 2022]. Available from: <https://www.tga.gov.au/resources/resource/guidance/australian-regulatoryguidelines-medical-devices-argmd>.

23. Australian Commission on Safety and Quality on Health Care. National Standard for User-applied Labelling of Injectable Medicines Fluids and Lines: ACSQHC; 2015 [23 August 2022]. Available from: <https://www.safetyandquality.gov.au/our-work/medicationsafety/safer-naming-and-labelling-medicines/national-standard-user-applied-labelling-injectable-medicines-fluids-and-lines>.
24. Department of Health. Clinical Incident Management Policy: Government of Western Australia; 2019 [15 September 2022]. Available from: <https://ww2.health.wa.gov.au/About-22> Guideline: Insertion and Management of Peripheral Intravenous Cannulae in Healthcare Facilities
25. Ponnusamy V, Venkatesh V, Clarke P. Skin antisepsis in the neonate: what should we use? *Curr Opin Infect Dis*. 2014;27(3):244-50.
26. Sharma A, Kulkarni S, Thukral A, Sankar MJ, Agarwal R, Deorari AK, et al. Aqueous chlorhexidine 1% versus 2% for neonatal skin antisepsis: a randomised noninferiority trial. *Arch Dis Child Fetal Neonatal Ed*. 2021;106(6):643-8.
27. Carbajal R, Veerapen S, Couderc S, Jugie M, Ville Y. Analgesic effect of breast feeding in term neonates: randomised controlled trial. *Bmj*. 2003;326(7379):13.
28. Yamada J, Stinson J, Lamba J, Dickson A, McGrath PJ, Stevens B. A Review of Systematic Reviews on Pain Interventions in Hospitalized Infants. *Pain Research and Management*. 2008;13:232316.
29. Child and Adolescent Health Service Neonatology. Clinical Guideline; Pain Assessment and Management. 2020.
30. Child and Adolescent Health Service Neonatology. Guideline; Medication Administration. 2022.
31. Child and Adolescent Health Service Neonatology. Guideline; Blood Sampling: Capillary, Venepuncture, Peripheral Arterial, UAC, UVC and CVC. 2019.
32. Ullman AJ, Cooke ML, Gillies D, Marsh NM, Daud A, McGrail MR, et al. Optimal timing for intravascular administration set replacement. *Cochrane Database Syst Rev*. 2013;2013(9):Cd003588.

18. Glossary

The following definitions are relevant to this procedure:

Term	Definition
PIVC	Peripherally Inserted Venous Cannulae
HCW	Health Care Worker

19. Document contact

Enquiries relating to this procedure may be directed to:

Title: Clinical Nurse Specialist, Vascular Access
 Department: Vascular Access Service
 Email: andries.niemann@health.wa.gov.au

20. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

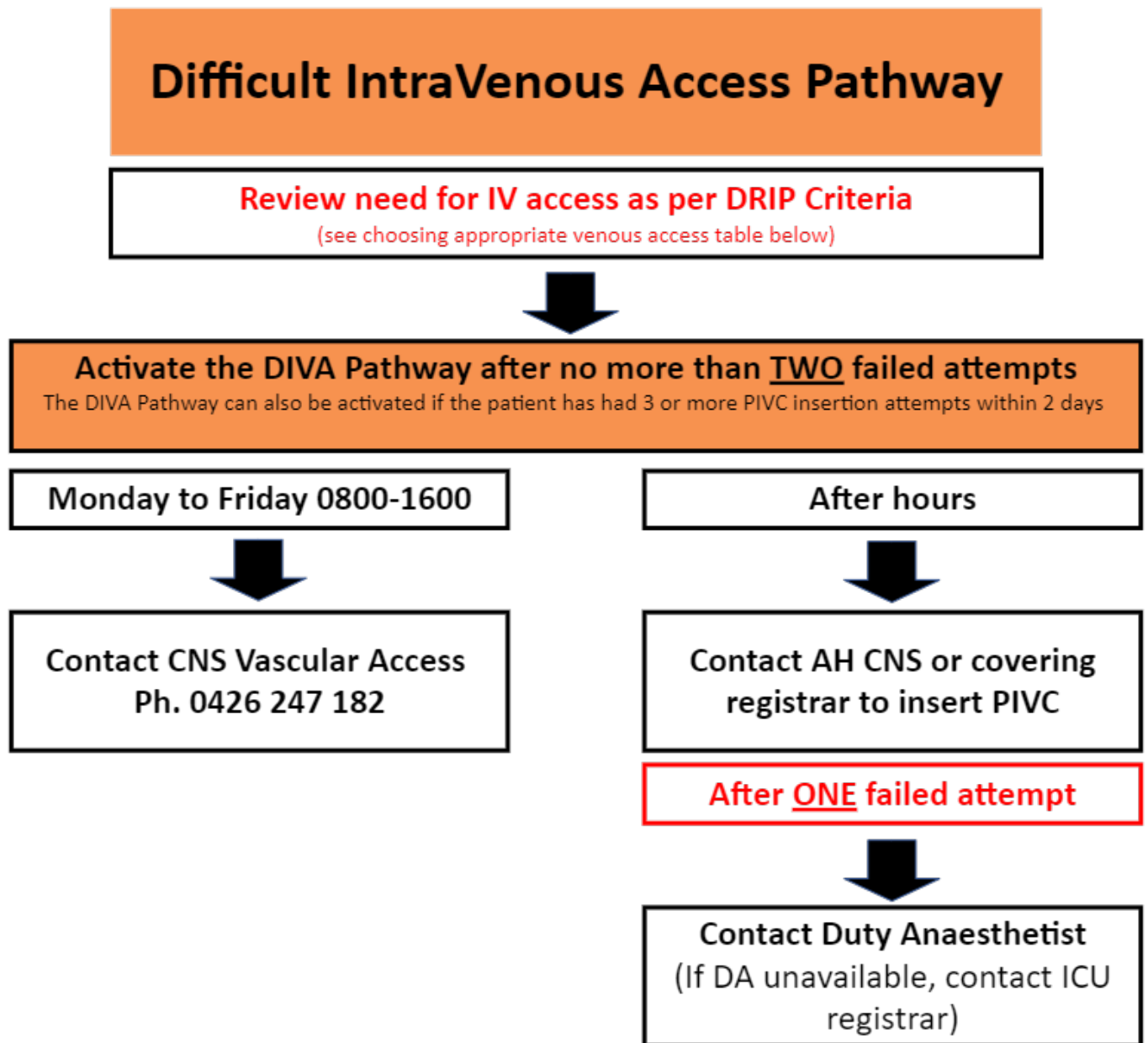
Version	Effective from	Effective to	Amendment(s)
V1.4	11/08/2025	17/01/2026	Minor amendment Section 9: Duration and re-sitting of PIVCs If the reviewing doctor decides that the PIVC MUST stay in the cubital fossa, this must be documented in the DMR and noted as discussed with a registrar as a minimum regarding infection risk associated with leaving it in place

21. Authorisation

This procedure has been authorised and issued by:

Authorised by	Jane Husain – A/Director Nursing and Midwifery
Authorisation date	06/08/2025
Published date	11/08/2025

Appendix 1 – Difficult Intravenous Access Pathway



If three or more attempts at inserting PIVC within 2 days, consider insertion of midline PICC.

Choosing appropriate venous access on the ward

A PIVC is only to be inserted if deemed clinically necessary and alternatives are not viable e.g. oral medications

Peripheral IntraVenous Cannulae (PIVC)	Indications: - Treatment involves infusion of peripherally compatible medications and solutions (less than 600mOsm/L, not and irritant or vesicant).
	Considerations: - Must be re-sited after 72 hours or earlier if indicated
Midline (Long Dwell Peripheral Cannula)	Indications: - IV therapy duration >5 days OR - Difficult IV access despite attempts at ultrasound guided PIVC insertion.
	Considerations: - Can remain in situ for up to 28 days - Not suitable for TPN, vesicant medications, or inotropes
Peripherally Inserted Central Catheters (PICC)	Indications: - IV access required for longer than 14 days - IV therapy with peripherally incompatible solutions
	Considerations: - Chest x-ray may be required, tip terminates at superior vena cava - Suitable for TPN, IV contrast, vesicants and inotropes - Single lumen catheters preferred unless specific indication for additional lumen - Procedure consent form must be obtained from patient prior to commencement of procedure

Appendix 2 – PIVC selection poster

PIVC SIZE	COLOUR	RATIONALE FOR USE
14G	ORANGE	Trauma patients Rapid, large-volume replacement
16G	GREY	High volume of fluids / Trauma patients Major surgery Intra-partum or post-partum / GIT Bleeding Multiple line access Multiple blood transfers
18G	GREEN	Rapid administration of large volumes Blood products / Viscous fluid infusions Delivery of irritant medications Multiple line access Major surgery Imaging requiring power injection of CT kontras
20G	PINK	General use / IV maintenance IV antibiotics IV analgesia
22G	BLUE	Small or fragile veins Paediatric Most types of drug therapy - continuous intermittent or bolus Cytotoxic therapy
24G	YELLOW	Small veins For slow flow rates Neonatal Cancer services

Selecting an appropriate site

- Consider the length of PIVC
- Start distally in the upper extremities
- Avoid antecubital fossa if possible
- Choose firm, round, elastic, well filled veins
- Assess the length of the vein
- Inspect and palpate for problems
- Look at or ask the patient for their previous history of cannulation (if possible)

Appendix 3 – Peripheral intravenous assessment score (PIVAS)

Assess the PIVC site each time it is accessed and ensure a PIVAS is documented each shift

LOOK	LISTEN	FEEL
Observe the PIVC site for erythema, swelling or exudate. Is the dressing intact, clean and dry?	Ask the patient or use visual clues. Is there pain or tenderness on infusion / palpation or movement?	Palpate the site through the intact dressing. Is there any heat or vessel hardening?

PIVAS CLINICAL ASSESSMENT AND INTERVENTIONS Always use Look , Listen and Feel observations (noted above)	
0	Healthy IV site - No signs of phlebitis - No identified concerns - Replace dressing if not clean, dry, or intact
1	One of the following is evident: Pain, tenderness or erythema at IV site - Discuss with Medical Officer and consider review of infusion rate or further dilution of medications - Replace dressing if not clean, dry and intact Continue to observe site closely and document each shift
2	Two of the following signs or symptoms are evident: Pain, erythema, swelling, discharge or palpable venous cord Remove PIVC immediately Send cannula tip for culture with accompanying blood cultures - Inform Medical Officer and re-site only if required - Document signs and symptoms, PIVAS and actions in patient's medical record - Complete clinical incident notification - Continue to observe and record status of IV site until healed
A PIVAS of 2 or more OR fever not explained by other causes requires collection of 2 sets of blood cultures and the PIVC tip sent for culture	
3	All of the following are evident: Pain along path of cannula, erythema, induration and palpable venous cord. Also possibly evident: Pus, pyrexia Remove PIVC immediately and inform Medical Officer - If ongoing IV treatment required consider alternate venous access device e.g. PICC - Document signs and symptoms, PIVAS and actions in patient medical record - Initiate additional treatment as required - Complete clinical incident notification - Continue to observe and record status of IV site until healed - If discharged from hospital advise GP for review

Adapted From: Jackson A (1998) *Infection control: a battle in vein infusion phlebitis*. *Nursing Times*; 94: 4, 68-71